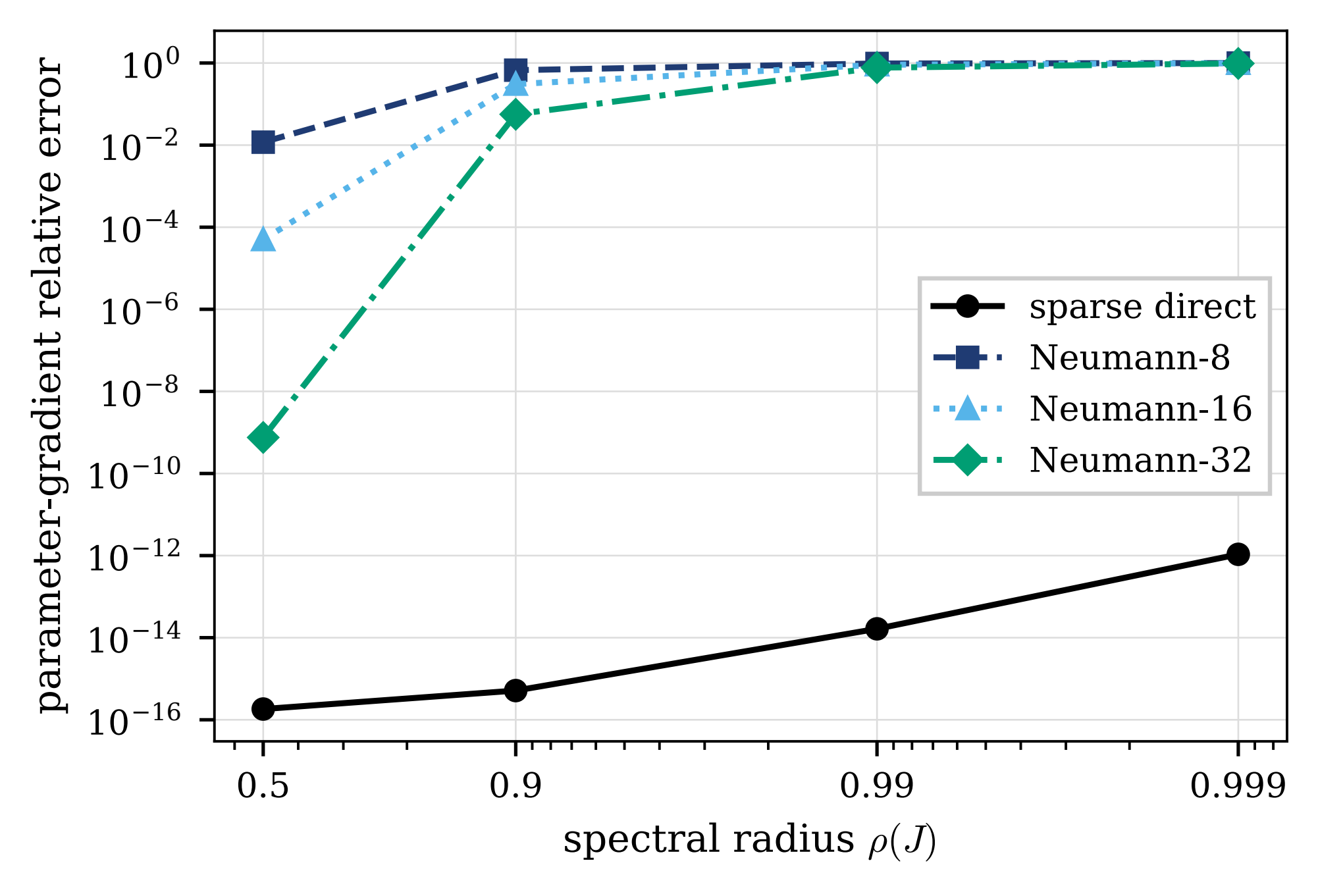
parameter-gradient relative error
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
0.5
0.9
0.99
0.999
spectral radius $\rho(J)$
sparse direct
Neumann-8
Neumann-16
Neumann-32